

3C: Full model

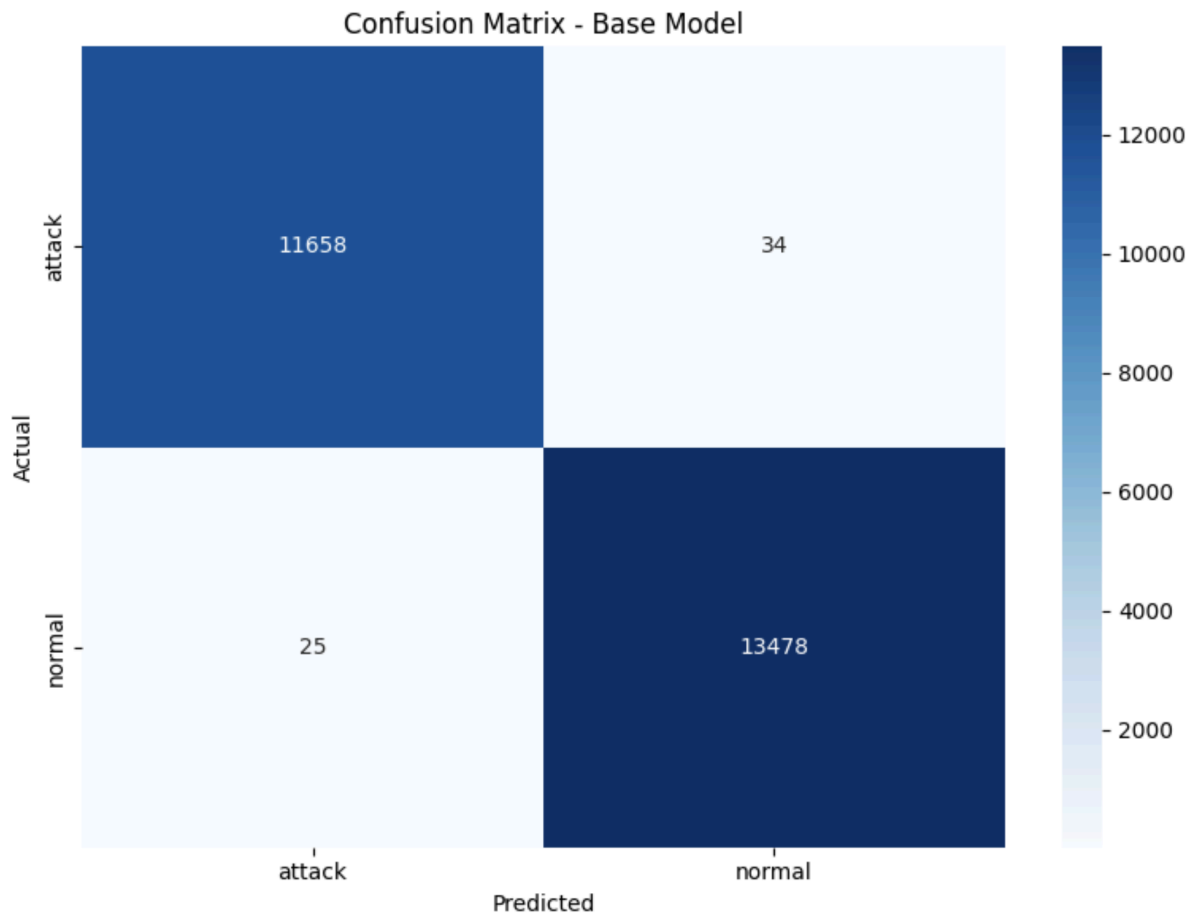
788/788 [=====] - 1s 674us/step

Classification Report - Base Model:

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11692
normal	1.00	1.00	1.00	13503
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix - Base Model:

```
[[11658   34]
 [   25 13478]]
```



4B: Quantized model

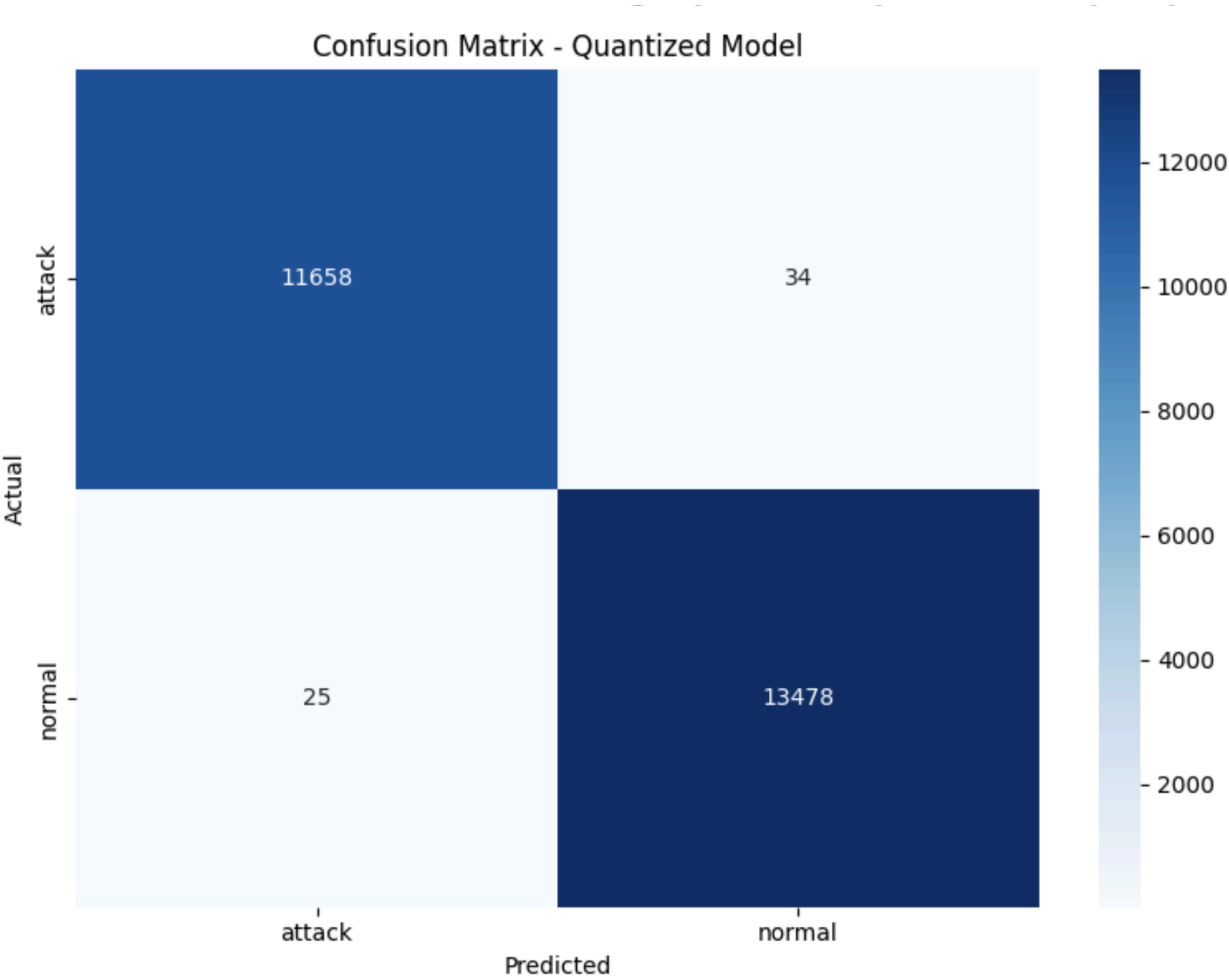
Classification Report – Quantized Model:

	precision	recall	f1-score	support
attack	1.00	1.00	1.00	11692
normal	1.00	1.00	1.00	13503
accuracy			1.00	25195
macro avg	1.00	1.00	1.00	25195
weighted avg	1.00	1.00	1.00	25195

Confusion Matrix – Quantized Model:

[[11658 34]

[25 13478]]



5C:

The screenshot shows the Arduino IDE with the file `network_data.ino` open. The code includes TensorFlow Lite headers and defines constants for the number of inputs, outputs, arena size, and samples. It then defines a `uint8_t` array `tensor_arena` and a `const float` array `X_test_5` with 38 elements. The Serial Monitor output shows the results of the model's predictions for 10 samples.

```
network_data.ino network_model.h
// TensorFlow Lite microcontroller interface
#include "tensorflow/lite/micro/micro_mutable_op_resolver.h"
#include "tensorflow/lite/schema/schema_generated.h"
#include "tensorflow/lite/version.h"

#define NUMBER_OF_INPUTS 38
#define NUMBER_OF_OUTPUTS 2
#define TENSOR_ARENA_SIZE 16*1024
#define NUM_SAMPLES 5

uint8_t tensor_arena[TENSOR_ARENA_SIZE];
tflite::ErrorReporter* error_reporter;
tflite::MicroInterpreter* interpreter;
TfLiteTensor* input;
TfLiteTensor* output;

const float X_test_5[38] = {
-0.11024922321249885, 2.219311849644404, -1.1761894520278895,
0.7511112872365361, -0.0077561081831758055, -0.004918644383724874,
-0.00948642202040107, -0.09507567152556495, -0.8092618187059747,
-0.011663642603760032, -0.036651869142258646, -0.024436507262009306,
-0.01238515036740332, -0.02618002418454278, -0.018609896340735923,
-0.04122119759327531, -0.0028174939213690777, -0.0975309439715147,
-0.7170454919159579, -0.32680812408381416, -0.6372092679572258,
```

Serial Monitor Output:

Sample #	Predicted Class	Actual Class
Sample #1	normal	normal
Sample #2	attack	attack
Sample #3	attack	attack
Sample #4	attack	attack
Sample #5	attack	attack
Sample #1	normal	normal
Sample #2	attack	attack
Sample #3	attack	attack
Sample #4	attack	attack
Sample #5	attack	attack

5D:

The screenshot shows the Arduino IDE with the file `network_data.ino` open. The code includes TensorFlow Lite headers and defines constants for the number of inputs, outputs, arena size, and samples. It then defines a `uint8_t` array `tensor_arena` and a `const float` array `X_test` with 38 elements. The Serial Monitor output shows the results of the model's predictions for 10 samples.

```
network_data.ino network_model.h
// TensorFlow Lite microcontroller interface
#include "tensorflow/lite/micro/micro_mutable_op_resolver.h"
#include "tensorflow/lite/schema/schema_generated.h"
#include "tensorflow/lite/version.h"

#define NUMBER_OF_INPUTS 38
#define NUMBER_OF_OUTPUTS 2
#define TENSOR_ARENA_SIZE 16*1024
#define NUM_SAMPLES 5

uint8_t tensor_arena[TENSOR_ARENA_SIZE];
tflite::ErrorReporter* error_reporter;
tflite::MicroInterpreter* interpreter;
TfLiteTensor* input;
TfLiteTensor* output;

const float X_test[NUM_SAMPLES][NUMBER_OF_INPUTS] = {
-0.11024922321249885, -0.12470615670462065, -0.44208308523109213,
0.7511112872365361, -0.007706877375215699, -0.0037565688369884287,
-0.00948642202040107, -0.09507567152556495, 1.2356940323701657,
-0.011663642603760032, -0.036651869142258646, -0.024436507262009306,
-0.01238515036740332, -0.02618002418454278, -0.018609896340735923,
-0.04122119759327531, -0.0028174939213690777, -0.0975309439715147,
-0.5598514445843843, -0.10653031654754139, -0.6372092679572258,
-0.6319290328885425, -0.37436223991967527, -0.37443160310530493,
0.7712831058493207, -0.349683038873482, -0.3745597040553465,
0.7343425609306344, 1.2587542737799418, 1.0664013456654926,
-0.43907816809041417, -0.4801968475158174, -0.28910340026287856,
-0.6395319051152512, -0.6248707997445304, -0.38763462350750655,
-0.3763870260600415, 0.6528228780141483, -0.11024922321249885,
-0.12470615670462065, 1.0873051789289023, -0.7362346401101137,
-0.00776224074056876, -0.004918644383724874, -0.00948642202040107,
-0.09507567152556495, -0.8092618187059747, -0.011663642603760032,
-0.036651869142258646, -0.024436507262009306, -0.01238515036740332,
```

Serial Monitor Output:

Sample #	Predicted Class	Actual Class
Sample #1	normal	normal
Sample #2	attack	attack
Sample #3	attack	attack
Sample #4	attack	attack
Sample #5	normal	normal
Sample #6	attack	attack
Sample #7	normal	normal
Sample #8	attack	attack
Sample #9	normal	normal
Sample #10	normal	normal
Sample #1	normal	normal
Sample #2	attack	attack